

Cooked meat and risk of breast cancer--lifetime versus recent dietary intake.



BACKGROUND: Polycyclic aromatic hydrocarbons (PAHs) and heterocyclic amines (HCAs) are carcinogens formed in or on the surface of well-done meat, cooked at high temperature. METHODS: We estimated breast cancer risk in relation to intake of cooked meat in a population-based, case-control study (1508 cases and 1556 controls) conducted in Long Island, NY from 1996 to 1997. Lifetime intakes of grilled or barbecued and smoked meats were derived from the interviewer-administered questionnaire data. Dietary intakes of PAH and HCA were derived from the self-administered modified Block food frequency questionnaire of intake 1 year before reference date. Unconditional logistic regression was used to estimate adjusted odds ratios (ORs) and 95% confidence intervals (CIs). RESULTS: Modest increased risk was observed among postmenopausal, but not premenopausal, women consuming the most grilled or barbecued and smoked meats over the life course (OR = 1.47; CI = 1.12-1.92 for highest vs. lowest tertile of intake). Postmenopausal women with low fruit and vegetable intake, but high lifetime intake of grilled or barbecued and smoked meats, had a higher OR of 1.74 (CI = 1.20-2.50). No associations were observed with the food frequency questionnaire-derived intake measures of PAHs and HCAs, with the possible exception of benzo(alpha)pyrene from meat among postmenopausal women whose tumors were positive for both estrogen receptors and progesterone receptors (OR = 1.47; CI = 0.99-2.19). CONCLUSIONS: These results support the accumulating evidence that consumption of meats cooked by methods that promote carcinogen formation may increase risk of postmenopausal breast cancer.